# آمار و احتمال مهندسی

# Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 15

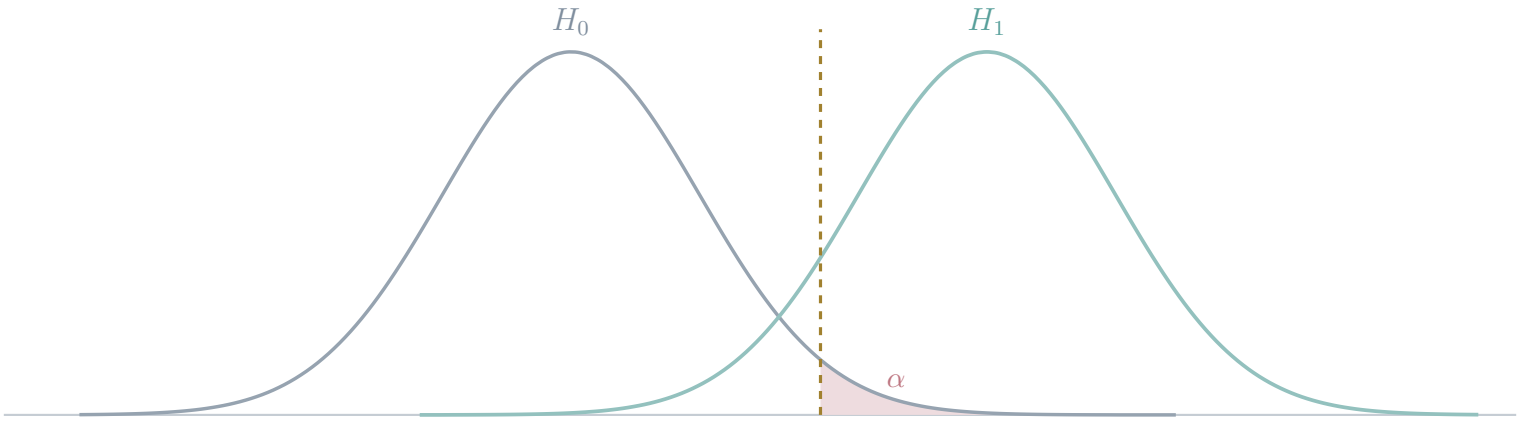
## Hypothesis Testing

آزمون فرض

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## Learning Objectives

After completing this module, students will be able to:

1. Formulate null and alternative hypotheses for engineering problems
2. Identify Type I and Type II errors and their engineering consequences
3. Compute test statistics and p-values
4. Perform Z-tests and t-tests for means
5. Correctly interpret p-values (and avoid common misconceptions)
6. Choose between one-sided and two-sided tests

## 15.1 Introduction: Making Decisions from Data

**Engineering question:** “Is this sensor biased?” “Did the process change?” “Does the new algorithm perform better?”

We need a **systematic framework** to answer such questions from data, while controlling the probability of making errors.

## 15.2 Framework

### Null Hypothesis $H_0$

The **default assumption** — typically "nothing changed" or "no effect":

- $H_0 : \mu = \mu_0$ (sensor is not biased)
- $H_0 : \mu_1 = \mu_2$ (two methods perform equally)
- $H_0 : \sigma^2 = \sigma_0^2$ (variance hasn't changed)

### Alternative Hypothesis $H_1$

What we want to **detect** or prove:

- $H_1 : \mu \neq \mu_0$ (sensor IS biased — two-sided)
- $H_1 : \mu > \mu_0$ (performance improved — one-sided)
- $H_1 : \sigma^2 > \sigma_0^2$ (variance increased)

### Decision

Based on data, either:

- **Reject** $H_0 \rightarrow$ evidence supports $H_1$
- **Fail to reject** $H_0 \rightarrow$ insufficient evidence against $H_0$

> ⚠ "Fail to reject $H_0$" $\neq$ "$H_0$ is true". Absence of evidence $\neq$ evidence of absence.

# 15.3 Types of Errors

| | $H_0$ True (reality) | $H_1$ True (reality) |
|---|---|---|
| **Reject** $H_0$ (decision) | Type I Error $(\alpha)$ | Correct! (Power $= 1 - \beta$) |
| **Fail to reject** $H_0$ (decision) | Correct! | Type II Error $(\beta)$ |

## Type I Error (False Alarm) — probability $\alpha$

Rejecting $H_0$ when it's actually true. "Crying wolf."

Engineering analogy: Declaring a process has changed when it hasn't $\rightarrow$ unnecessary recalibration.

## Type II Error (Missed Detection) — probability $\beta$

Failing to reject $H_0$ when $H_1$ is actually true. "Missing the signal."

Engineering analogy: Failing to detect a sensor bias $\rightarrow$ biased measurements continue.

## Engineering Analogy: Radar Detection

| Statistical Term | Radar Equivalent |
|---|---|
| $H_0$: no target | No aircraft present |
| $H_1$: target present | Aircraft present |
| Type I error $(\alpha)$ | False alarm |
| Type II error $(\beta)$ | Missed detection |
| Power $(1 - \beta)$ | Probability of detection |

## The Tradeoff

Reducing $\alpha$ (fewer false alarms) $\rightarrow$ increases $\beta$ (more missed detections), and vice versa.

Fix $\alpha$ (significance level), then maximize power $= 1 - \beta$.

## 15.4 Significance Level $\alpha$

The **maximum acceptable Type I error rate**, chosen BEFORE looking at data.

Common choices: $\alpha = 0.05,\ 0.01,\ 0.10$

**Choosing $\alpha$:** Consider the cost of a false alarm:

- $\alpha = 0.01$: Very conservative (e.g., medical device safety)
- $\alpha = 0.05$: Standard in most engineering/science
- $\alpha = 0.10$: More tolerant (exploratory analysis)

## 15.5 Power of a Test

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 | H_1 \text{ is true})$$

Power depends on:

1. **Effect size:** Larger true difference $\rightarrow$ easier to detect $\rightarrow$ more power
2. **Sample size $n$:** More data $\rightarrow$ more power
3. **Significance level $\alpha$:** Larger $\alpha \rightarrow$ more power (but more false alarms)
4. **Variability $\sigma$:** Less noise $\rightarrow$ more power

## 15.6 The p-value

**Definition**

The **p-value** is the probability of obtaining a test statistic **as extreme or more extreme** than the observed value, **assuming $H_0$ is true**.

$$p\text{-value} = P(\text{data this extreme or more} | H_0 \text{ true})$$

### Decision Rule

- If p-value $< \alpha \rightarrow$ Reject $H_0$
- If p-value $\geq \alpha \rightarrow$ Fail to reject $H_0$

### ✔ CORRECT Interpretation

> "The p-value is the probability of seeing data this extreme (or more) if $H_0$ were true. A small p-value means the data is unlikely under $H_0$."

### ✘ WRONG Interpretation

> ~~ "The p-value is the probability that $H_0$ is true." ~~

**Why wrong:** $H_0$ is either true or false (fixed). The p-value says nothing about $P(H_0 \text{ true})$. It measures how surprising the data is under $H_0$.

### Calibration

| p-value | Evidence against $H_0$ |
|---|---|
| $> 0.10$ | Weak or none |
| 0.05 – 0.10 | Marginal |
| 0.01 – 0.05 | Moderate |
| 0.001 – 0.01 | Strong |
| $< 0.001$ | Very strong |

## 15.7 One-Sided vs. Two-Sided Tests

### Two-Sided ($H_1$: $\mu \neq \mu_0$)

Used when deviation in **either** direction is important.

- Reject if $|T| > t_{\alpha/2,n-1}$
- p-value $= 2 \cdot P(T > |t_{\text{obs}}|)$

### One-Sided ($H_1$: $\mu > \mu_0$ or $H_1 : \mu < \mu_0$)

Used when only one direction of deviation matters.

- $H_1 : \mu > \mu_0$: Reject if $T > t_{\alpha,n-1}$
- $H_1 : \mu < \mu_0$: Reject if $T < -t_{\alpha,n-1}$

### When to Use Which

| Scenario | Test Type |
|---|---|
| Is the sensor biased (either direction)? | Two-sided |
| Does the new algorithm IMPROVE BER? | One-sided ($H_1$: BER $<$ BER$_{\text{old}}$) |
| Has noise power INCREASED? | One-sided ($H_1$: $\sigma^2 > \sigma_0^2$) |
| Did the process mean CHANGE? | Two-sided |

## 15.8 Tests for Means

### One-Sample Z-Test ($\sigma$ known)

Test: $H_0 : \mu = \mu_0$

Test statistic: $Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$

Under $H_0 : Z \sim N(0,\, 1)$

### One-Sample t-Test ($\sigma$ unknown)

Test: $H_0 : \mu = \mu_0$

Test statistic: $T = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$

Under $H_0 : T \sim t(n-1)$

### Two-Sample t-Test (comparing two means)

Test: $H_0 : \mu_1 = \mu_2$

Test statistic: $T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$

Approximate $df$ by Welch-Satterthwaite formula.

# 15.9 Test for Variance (Chi-Square Test)

## One-Sample Test

$H_0 : \sigma^2 = \sigma_0^2$

Test statistic: $\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$

Under $H_0 : \chi^2 \sim \chi^2(n-1)$

## Decision

- $H_1 : \sigma^2 > \sigma_0^2$: Reject if $\chi^2 > \chi^2_{\alpha,n-1}$
- $H_1 : \sigma^2 \neq \sigma_0^2$: Reject if $\chi^2 < \chi^2_{1-\alpha/2,n-1}$ or $\chi^2 > \chi^2_{\alpha/2,n-1}$

## 15.10 Step-by-Step Hypothesis Testing Procedure

1. **State hypotheses:** $H_0$ and $H_1$
2. **Choose significance level:** $\alpha$ (e.g., 0.05)
3. **Select test statistic:** $Z$, $t$, or $\chi^2$ depending on scenario
4. **Determine critical value(s)** or compute p-value
5. **Compute test statistic** from data
6. **Make decision:** Reject $H_0$ if test statistic in critical region (or $p < \alpha$)
7. **State conclusion** in engineering terms

## 15.11 Engineering Examples

### Example 1: Is a Sensor Biased?

A sensor should read 0V with no input. 20 measurements give $\bar{X} = 0.012\,\text{V}$, $S = 0.03\,\text{V}$.
$H_0 : \mu = 0$ (no bias), $H_1 : \mu \neq 0$ (biased), $\alpha = 0.05$

$$T = \frac{0.012 - 0}{\frac{0.03}{\sqrt{20}}} = \frac{0.012}{0.00671} = 1.789$$

$t_{0.025,19} = 2.093$. Since $|T| = 1.789 < 2.093 \rightarrow$ **Fail to reject $H_0$.**
p-value $= 2 \cdot P(t(19) > 1.789) \approx 0.090$. Not significant at 5% level.
Conclusion: Insufficient evidence to conclude the sensor is biased.

### Example 2: Did Manufacturing Process Change?

Before: $\mu_0 = 100\,\Omega$ (established). After modification: $n = 30$, $\bar{X} = 101.2\,\Omega$, $S = 3.5\,\Omega$.
$H_0 : \mu = 100$, $H_1 : \mu \neq 100$, $\alpha = 0.05$

$$T = \frac{101.2 - 100}{\frac{3.5}{\sqrt{30}}} = \frac{1.2}{0.639} = 1.878$$

$t_{0.025,29} = 2.045$. $|T| = 1.878 < 2.045 \rightarrow$ **Fail to reject $H_0$.**

### Example 3: Does New Algorithm Improve BER?

Old algorithm: $\text{BER}_0 = 10^{-3}$. New algorithm tested: $n = 50000$ bits, 38 errors.

$$\text{BER}_{\text{new}} = \frac{38}{50000} = 7.6 \times 10^{-4}$$

$H_0 : p = 0.001$, $H_1 : p < 0.001$ (improvement), $\alpha = 0.05$

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.00076 - 0.001}{\sqrt{0.001 \times \frac{0.999}{50000}}} = -\frac{0.00024}{0.000141} = -1.70$$

p-value $= P(Z < -1.70) = 0.0446 < 0.05 \rightarrow$ **Reject $H_0$.**
Conclusion: Evidence supports that the new algorithm has lower BER.

### Example 4: Has Noise Power Increased?

Historical: $\sigma_0^2 = 0.04\,\text{V}^2$. New measurements: $n = 25$, $S^2 = 0.058\,\text{V}^2$.
$H_0 : \sigma^2 = 0.04$, $H_1 : \sigma^2 > 0.04$, $\alpha = 0.05$

$$\chi^2 = \frac{(24)(0.058)}{0.04} = 34.8$$

$\chi^2_{0.05,24} = 36.415$. Since $34.8 < 36.415 \rightarrow$ **Fail to reject $H_0$.**

## 15.12 Practice Problems

**Problem 1**

A manufacturing spec requires mean resistance = 100Ω. A sample of 16 resistors gives $\bar{X} = 101.5\,\Omega$, $S = 3\,\Omega$. At $\alpha = 0.05$, is there evidence the process has drifted?

**Solution** $H_0 : \mu = 100$, $H_1 : \mu \neq 100$. $T = \frac{101.5-100}{\frac{3}{4}} = 2.0$. $t_{0.025,15} = 2.131$. $|T| = 2.0 < 2.131$ $\rightarrow$ Fail to reject. p-value $\approx 0.064 > 0.05$.

**Problem 2**

Average packet delay was 5ms. After network upgrade, 40 measurements give $\bar{X} = 4.6\,\text{ms}$, $S = 1.2\,\text{ms}$. Test if delay decreased at $\alpha = 0.01$.

**Solution** $H_0 : \mu = 5$, $H_1 : \mu < 5$. $T = \frac{4.6-5}{\frac{1.2}{\sqrt{40}}} = -\frac{0.4}{0.190} = -2.11$. $t_{0.01,39} \approx -2.426$. $T = -2.11 > -2.426 \rightarrow$ Fail to reject at 1% level. (Would reject at 5% since $t_{0.05,39} \approx -1.685$.)

**Problem 3**

Explain why "$p = 0.04$ means there is only a 4% chance $H_0$ is true" is WRONG.

**Solution** The p-value is $P(\text{data this extreme} \mid H_0 \text{ true})$, NOT $P(H_0 \text{ true} \mid \text{data})$. It conditions on $H_0$ being true and asks about data extremity. To get $P(H_0 \text{ true} \mid \text{data})$ would require Bayesian analysis with a prior. The p-value is a property of the data-generating process under $H_0$, not a posterior probability about $H_0$.

*Next Module: Module 16 — Engineering Statistical Applications*